A splitting approach for variational inference in non-Gaussian latent Gaussian models

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Abstract

Many analyses of high-dimensional biological data begin by transforming non-Gaussian observations and then applying a Gaussian method. Examples include log-transforming RNA sequencing counts before factor analysis and transforming accessibility proportions before functional regression. These transformations are convenient, but they discard information about the observation process and can be sensitive to arbitrary choices such as a pseudocount. We develop a modular variational framework that couples an exponential-family observation model to an existing empirical Bayes Gaussian method through a latent Gaussian splitting variable. The algorithm alternates between independent low-dimensional variational updates for the non-Gaussian observations and a Gaussian update delegated to software for normal means, regression, smoothing, matrix factorization, multivariate analysis, or structured variable selection.

We give explicit updates for Poisson and binomial observations. The binomial case is especially useful when a response aggregates binary measurements: the number of successes and trials are retained, so observations with different sampling depths contribute different information. We clarify an important limitation. With one Bernoulli observation per latent mean, the splitting variance can be weakly identified or non-identifiable; replication within a binomial observation supplies extra-binomial information. We conclude with a binomial functional regression model for single-molecule chromatin accessibility data, using wavelet-based functional fine-mapping on the latent log-odds scale.

1. Introduction

Consider observations y_i whose distribution belongs to an exponential family, with latent canonical or transformed mean mu_i. Depending on the structure assigned to the joint prior of the latent means, this formulation includes empirical Bayes normal means, sparse regression, smoothing, multivariate shrinkage, and matrix factorization. The Gaussian versions of these problems are computationally mature, whereas replacing Gaussian observations by counts or proportions often requires a new inference algorithm for every structured prior.

$$y_i \mid mu_i \sim D\{h(mu_i)\}, \quad mu \sim f. \qquad (1)$$

Our goal is to reuse Gaussian methods without first transforming the data. We introduce a latent Gaussian layer that separates the likelihood from the structured prior. One module handles the non-Gaussian likelihood; the other calls an empirical Bayes Gaussian method (EBG). This is analogous in spirit to splitting methods in optimization, where a coupled problem is replaced by two simpler problems connected through low-dimensional posterior moments.

The original motivation was dimension reduction for single-cell RNA sequencing counts. A second important use arises when binary measurements are aggregated within a biological sample. In Fiber-seq, accessibility is observed on individual chromatin fibers but genetic covariates are measured at the donor level. The relevant observation is a number of accessible fibers out of a known number assayed. A proportion alone cannot distinguish one out of two from one hundred out of two hundred, and treating fibers as independent genetic samples would be pseudoreplication.

Table 1. Examples of reusable Gaussian modules.

Structure for b	Model	Gaussian engine
b_i iid ~ g	univariate mean inference	ash, ebnm
b = X beta, beta ~ g	regression	sparse EB regression
b spatially smooth	smoothing	wavelet or GP EB
b = L F'	matrix factorization	flash
b_i ~ mixture N(0,U_k)	multivariate means	mash
b = X B, structured B	functional selection	fSuSiE

2. Split variational inference

2.1 The augmented model

Introduce a splitting variable b and write

$$y_i \mid mu_i \sim D\{h(mu_i)\},$$
$$mu_i \mid b_i, tau_i^2 \sim N(b_i, tau_i^2), \quad b \sim g. \qquad (2)$$

The splitting variance may be an algorithmic device connecting the likelihood to a Gaussian engine, or part of the scientific model representing residual variation on the latent scale. This distinction matters for identifiability. We use q(mu,b) = {product_i q(mu_i)} q(b), with each local factor q(mu_i) = N(m_i,v_i). Let bbar_i and Vb_i denote the posterior mean and variance returned by the Gaussian module.

$$F(q,g;tau^2) = sum_i E \log p(y_i|mu_i) + sum_i E \log p(mu_i|b_i,tau_i^2) + E \log\{g(b)/q(b)\} - sum_i E \log q(mu_i). \qquad (3)$$

2.2 Modular coordinate-ascent algorithm

1. **Non-Gaussian step.** Update every $q(mu_i)$ using its likelihood, current prior mean $bbar_i$, and splitting variance tau^2.
2. **Gaussian step.** Treat $m = (m_1,...,m_M)$ as a Gaussian response with residual variance tau^2; update $q(b)$ and the hyperparameters of g with EBG.
3. **Variance step.** If the splitting variance is estimated, include uncertainty from both variational blocks in its M-step.

$$tau^2 \leftarrow (1/M)\ sum_i \{(m_i\text{-}bbar_i)^2 + v_i + Vb_i\}. \quad (4)$$

The term Vb_i is essential. Omitting it generally underestimates latent residual variation. The same update applies entrywise to matrices. Known heterogeneous splitting variances can also be used, but then the Gaussian engine must accept heterogeneous standard errors.

2.3 Relation to a collapsed objective

Let $f_{tau}(mu)$ be the Gaussian convolution of a prior G. Its log curvature has a simple lower bound:

$$d^2 \log f_{tau}(mu) / dmu^2 = -1/tau^2 + Var(b|mu)/tau^4 \geq -1/tau^2. \quad (5)$$

For $q(mu)=N(m,v)$, this gives $E \log f_{tau}(mu)$ at least $\log f_{tau}(m) - v/(2\ tau^2)$. The right-hand side is exactly the result of profiling the splitting ELBO over $q(b)$. Thus the split objective is a lower bound on the variational objective obtained after integrating out b. Equality holds when G is a point mass; with a flexible G, the gap is the price of modularity.

3. Variational Gaussian updates

3.1 Poisson observations

For $y_i|mu_i \sim Poisson\{s_i\ exp(mu_i)\}$, the local objective for $q(mu_i)=N(m_i,v_i)$ is

$$F_i = y_i m_i - s_i exp(m_i+v_i/2) - \{(m_i\text{-}bbar_i)^2+v_i\}/(2tau^2) + (1/2)log\ v_i. \quad (6)$$

The negative objective is strictly convex. Its stationary conditions are

$$0 = y_i - s_i exp(m_i+v_i/2) - (m_i\text{-}bbar_i)/tau^2,$$
$$1/v_i = s_i exp(m_i+v_i/2) + 1/tau^2. \quad (7)$$

Damped Newton updates or a one-dimensional root solver are stable. The second condition implies $0 < v_i < tau^2$.

3.2 Binomial observations

Let $k_i|mu_i \sim Binomial\{n_i,logit^{-1}(mu_i)\}$, with $0 \le k_i \le n_i$. The local objective is

$$F_i = k_i m_i - n_i A(m_i,v_i) - \{(m_i\text{-}bbar_i)^2+v_i\}/(2tau^2) + (1/2)log\ v_i,$$
$$A(m,v) = E_{Z\sim N(m,v)}\ log(1+exp\ Z). \quad (8)$$

Gauss-Hermite quadrature accurately evaluates the logistic-normal expectation:

$$A(m,v) \approx (1/sqrt(pi))\ sum_{r=1}^{R}\ w_r\ log\{1+exp(m+sqrt(2v)\ z_r)\}. \quad (9)$$

Stable code uses a softplus evaluation. Exact derivative identities are $A_m = E\ p(Z)$ and $A_v = (1/2)E\ p(Z)\{1\text{-}p(Z)\}$, where $p(z)=logit^{-1}(z)$. The local first-order conditions are

$$0 = k_i - n_i E\, p(Z_i) - (m_i - bbar_i)/tau^2,$$
$$1/v_i = 1/tau^2 + n_i E[p(Z_i)\{1-p(Z_i)\}]. \quad (10)$$

We recommend optimization in $(m_i, \log v_i)$ with a damped Newton or trust-region method, using the same quadrature for the objective and derivatives. Warm starts make these independent two-parameter updates inexpensive.

3.3 Bernoulli identifiability and replication

The binomial extension requires more care than the Poisson extension. Suppose $x|mu \sim Bernoulli\{logit^{-1}(mu)\}$ and $mu \sim N(0,tau^2)$. Symmetry and $p(z)+p(-z)=1$ imply $Pr(x=1)=1/2$ for every tau^2. The Bernoulli likelihood therefore contains no information about the splitting variance in this case. Nonzero or varying prior means can supply information, but it may remain weak, and a flexible prior scale can be confounded with tau.

Replication within a binomial observation changes the variance:

$$Var(K) = n\, pbar(1-pbar) + n(n-1)\, Var(P), \quad pbar = E(P). \quad (11)$$

The extra-binomial term vanishes at $n=1$ and grows quadratically with n. Aggregated binomial data can therefore identify latent heterogeneity through repeated trials, whereas a single Bernoulli outcome cannot. Practical identification also depends on the number of independent biological units, range of trial totals, range of probabilities, and flexibility of the structured prior.

- Retain trial totals and diagnose their distribution instead of analyzing proportions alone.
- Compare fixed and estimated splitting variances; profile the objective for flat regions or boundary optima.
- Use null simulations or permutations that preserve trial totals and dependence structure.

3.4 Alternative augmentations for binary data

With one Bernoulli trial per latent mean, a fixed splitting variance can still be a computational approximation but should not be interpreted as estimated overdispersion. Albert-Chib augmentation is exact for a *probit* likelihood, $Pr(x=1)=Phi(b)$, not for the logistic sigmoid. Polya-Gamma augmentation is conditionally Gaussian for logistic binomial models. It produces observation-specific precisions, requiring a weighted Gaussian engine. Split VA trades this downstream complexity for independent local updates and a common latent residual variance.

4. Reusable Gaussian modules

4.1 Empirical Bayes normal means

When b_i are iid from g, the Gaussian step is the empirical Bayes normal means problem $m_i|b_i \sim N(b_i,tau^2)$. Scale mixtures of normals, unimodal families, or nonparametric priors can be fitted with existing software. Their posterior means and variances feed the next split-VA iteration.

4.2 Regression, smoothing, and multivariate means

For $b=X\,beta$, the Gaussian module is ordinary empirical Bayes regression. For a smooth latent function, it may be a Gaussian-process or wavelet smoother. For multivariate effects, it may fit a mixture $b_i \sim sum_k\ pi_k N(0,U_k)$. The Poisson or binomial observation code remains unchanged across these choices.

4.3 Matrix factorization

$$y_{ij}|mu_{ij} \sim D\{h(mu_{ij})\}, \quad mu_{ij}|L,F \sim N(sum_r\, l_{ir} f_{jr}, tau_{ij}^2). \quad (12)$$

The non-Gaussian module updates each matrix entry, while the Gaussian module fits factor analysis to the matrix of latent means $M=(m_{ij})$. Sparsity, non-negativity, or smoothness can be imposed through priors on loadings or factors without changing the observation module.

4.4 Binomial functional regression and Fiber-seq

Let A_{irt} be the binary accessibility call for fiber r from donor i at position t. Set $K_{it} = sum_r A_{irt}$, and let N_{it} be the number of observed fibers. A donor-level functional model is

$$K_{it}|mu_{it} \sim Binomial\{N_{it}, logit^{-1}(mu_{it})\},$$
$$mu_{it}|a_t,B,tau^2 \sim N\{a_t+(XB)_{it}, tau^2\}. \quad (13)$$

X contains donor-level genotypes and covariates, a_t is baseline accessibility, and rows of B are SNP effect curves on the log-odds scale. A functional Sum of Single Effects prior can be placed on B: each component selects one SNP, while a sparse wavelet prior regularizes its effect curve. The likelihood uses the binary sampling process, and the wavelet transform is applied to latent effect functions rather than directly to binary observations.

One binomial fSuSiE iteration would:

1. update every $q(mu_{it})$ using the binomial VGA with prior mean $a_t + E(XB)_{it}$ and variance tau^2;
2. fit Gaussian fSuSiE to $M=(m_{it})$, holding its residual variance fixed at tau^2;
3. update the baseline curve and the first two posterior moments of $a_t+(XB)_{it}$;
4. update tau^2 with posterior uncertainty from the latent and fSuSiE blocks.

The raw inputs must include both K_{it} and N_{it}. Supplying only their ratio loses sampling precision. If fibers are more correlated than the logistic-normal layer accommodates, donor- or region-specific random effects can be added. Validation should reproduce observed coverage, baseline accessibility, within-fiber spatial dependence, and linkage disequilibrium among SNPs.

5. Computation and diagnostics

The local Poisson and binomial updates are embarrassingly parallel. Their cost is linear in the number of observations times the number of quadrature nodes; the Gaussian module usually dominates running time. Recommended diagnostics include:

- monotonicity of the full split ELBO, evaluated with the same quadrature used for updates;
- relative changes in latent means, structured means, and the splitting variance;
- frequency of local solver failures and boundary values of v_i;
- profiles of the objective over tau^2;
- calibration under null simulations or covariate permutations;
- sensitivity to quadrature order, initialization, and fixed versus estimated tau^2.

For binomial functional regression, observed and fitted success counts should also be compared across coverage strata and positions. A method that fits mean accessibility but fails to reproduce donor-to-donor variance is not adequately modeling the experiment, even if its fitted mean curves look plausible.

6. Discussion

Split variational inference makes non-Gaussian modeling modular. The likelihood is handled by simple local variational problems, while structural learning is delegated to adaptive shrinkage, multivariate empirical Bayes, factor analysis, or functional fine-mapping. The binomial extension contributes both a useful application and a necessary warning: aggregated binary data with known totals are binomial counts whose precision depends on trial number and success probability, whereas pure Bernoulli data can provide little information about an extra latent variance.

For Fiber-seq fine-mapping, the binomial formulation is scientifically more defensible than applying a homoskedastic Gaussian model to mean accessibility. It is also computationally realistic because the Gaussian fSuSiE engine can be reused once it accepts a latent response and a fixed residual variance. The main research tasks are the binomial local solver, propagation of posterior uncertainty, identifiability diagnostics, and realistic calibration experiments rather than a redesign of functional fine-mapping.

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